WERGONIC · BACKEND · CLIENT PANEL · ADMIN · MOBILE

Open Work

Everything still open, re-read in the code today against the latest `dev` on all four repositories. Fifteen items from rev 4 are gone because they are fixed; they are listed at the end so nobody puts them back. Item numbers are unchanged from rev 4, so the two revisions can be compared side by side.

Rev 5 · 21 August 2026
Read against `dev`: backend `00c6e35` · web apps `cb2d905d` · mobile `96aed6a6` · landing `bb99528`

SINCE 12 AUGUST

Fifteen items closed, and one decision answered by the code

All three teams shipped. The head-backward threshold is now −5° everywhere — the risk verdict, the Word report, RAMP and the phone all agree — which answers DEC-1 without you having to. Trunk stays −10°, as intended.

The full list of what closed is on the last page.

READ BEFORE YOU ASSIGN

Two answers we were given do not match the code

From the RAMP/MEC answers on 18 August:

- **"Hand distance already takes the task into account."** It does not. The code that works out hand distance was rewritten on 18 August and still reads the whole recording — no task window is passed to it at all. **B4 stands.**
- **"MEC 12.1 and 12.2 removed."** Both are still in the backend's answer on `dev`, still empty on every MEC. Whatever was removed was not pushed. **B2 stands.**

MOVED, NOT DROPPED

The web speed items are not on this list any more

F1 to F6, F8 and F9 were handed to the web team on 17 August as *Web — speed cleanup*. They are live work, so they are out of this list rather than sitting in two places at once. F7, F10, F11 and F12 — the backend and phone ones — are still here.

A · Broken for users right now

All re-read in the code on 21 August, except where the item says otherwise.

A1 A measurement you retry from Home never comes back HIGH WEB

The Home list shows measurements that need attention. When you retry a failed one, it disappears from the list immediately — and the app remembers that it is hidden, in your browser, forever.
So if the retry then fails again, the measurement never reappears on Home. You will not be told and you will not see it there again. It is also hidden only on your computer, so a colleague still sees it as if nothing happened, and if you sign in from a different machine it is back.
An item should leave the Home list because the thing that put it there is resolved, not because someone clicked something. Work that out on the server, not in the browser.

A2 A manual RAMP you saved cannot be opened again HIGH WEB

Create a RAMP by hand and you are taken straight into the form. You can fill part of it and press Save. But once you go back to the assessments list, that assessment cannot be opened — clicking the row does nothing.
Your answers are saved and unreachable, so the only way forward is to delete it and start over. An unfinished manual RAMP must reopen its form when you click it, the same way RULA and REBA already do.

> **NOTE FOR YOU**
>
> The code says this out loud: the comment above the click handler reads "pure-manual RAMP have no click-through". It was written that way on purpose and never finished.

A5 Searching for a worker still misses part of an ID BACKEND

Half of this improved on its own. The Workers search now matches the start of any *word* in the ID, so typing `0451` finds `SE2026-0451`. What still fails is an ID with no separators: in `SE20260451`, only `SE` matches. Match anywhere in the ID.
Also still open, and worth doing at the same time: age accepts any number at all, so ages like 333 are in the data. Limit it to something sensible.

A7 Picking an activity on the phone looks like it worked when it did not MOBILE

Choosing an activity closes the picker whether or not the choice was saved — the failure is sent to Sentry and the screen closes anyway, so the user returns to the recording screen believing they picked something. The conditions screen is the same.
Keep the screen open and show the problem until it actually saves.

A11 The admin panel throws away what you typed WEB

A failed save closes the organisation form and loses everything entered. Filters break on names containing "&" or spaces. The organisation filter silently drops selections the user did not touch. And the counts do not refresh after adding a user.

> **NOTE FOR YOU**
>
> Carried over from 10 August and not re-read since — the admin app has had no commits touching these screens, so nothing can have changed, but nobody has opened them either.

A12 A recording of zero seconds is treated as a real measurement WEB BACKEND

A recording that captured nothing is marked complete, sits in the list like any other, and can be sent for a risk assessment. Nothing in the code checks the duration before doing any of that.
Mark it Failed instead, and do not offer it for assessment.

B · The numbers we report are wrong

A customer cannot tell any of these are wrong by looking. Re-read on 21 August.

B2 Two MEC questions report the wrong thing and two report nothing HIGH BACKEND

Questions 11.1 and 11.2 — the screen calls them "head in the red zone" — report the difference between the head and the trunk, which is what question 12 was meant to measure. Questions 12.1 and 12.2 are never worked out at all and come back empty on every MEC we have ever produced.
We were told 12.1 and 12.2 had been removed. They have not; both are still in the answer on `dev`.

> **NOTE FOR YOU**
> This one waits on DEC-6. The developer asked whether the title should change instead of the calculation, and that is your call, not theirs. Nothing here should be built until DEC-6 is answered.

B4 A RAMP for one task uses hand data from the whole recording HIGH BACKEND

Five of the six posture measures correctly look only at that task's time window. The sixth — how far the hands are from the body — uses the entire recording, so a result that is supposed to be about one task is mixed with the rest of the session.
The 18 August rewrite made this worse, not better: hand distance is now recalculated from the raw arm and forearm data so it can use the assessor's arm lengths, and that new calculation is given the session and nothing else — no task, no start time, no end time.

> **NOTE FOR YOU**
> We were told this was already handled because the shared filter function "takes the task into consideration". It can, but only when it is given one. Here it is not.

ID	WHERE	TASK
B6	Backend	In MEC question 11 the head and trunk readings are matched by their position in the list instead of by time. The two sensors record at different rates, so from the first dropped reading onwards it compares the head at one moment against the trunk at another, and the leftover readings are silently discarded. Everything else was already changed to match by time; this one place was missed. Ask why the earlier attempt at this was reverted before redoing it.
B7	Backend	The total time reported for a limb can be longer than the recording itself. Every raw read is bounded by the session window plus a deliberate one-day margin on each side, so a limb total can count readings from up to a day either side of the recording.
B8	Backend	The AI section can go missing from a report with nothing said. If the company has AI switched off, ticking the box is ignored. If the AI call fails, that is swallowed too. Either way the user is told the report worked and gets a document without the section they asked for. Waits on DEC-7.
B9	Backend	Six checkboxes in Generate report do nothing. Remove four of them — Worker details, Recorder details, Session duration, Session metadata — the report makes no sense without those tables, so they should not be optional. Make the other two, AI risk summary and Appendix, actually include or exclude their section.

C · Security — nothing is broken, but the server trusts the app

READ BEFORE ASSIGNING

None of these affect a user, and none will show up in testing

Our apps always send the right values, so the product works. The gap is that signing in proves who someone is, not what they are allowed to touch — so someone calling the API directly, outside our apps, can send something ours never would. There is no sign any of it has been used. Whether it is worth funding now is a business call: see DEC-3.

C1 The server decides whose data to return from the request, not from who is signed in HIGH BACKEND

Several Work Library endpoints — activities, conditions, and the labels put on a recording — pick which company's data to use from a number in the web address rather than from the signed-in user. Any valid login can read, create, edit or delete another company's data by sending a different number. Take the company from the signed-in user and ignore the one in the address.

> **NOTE FOR YOU**
>
> Six endpoints still work this way; four in the same area were already corrected, so there is a pattern to copy. This is not dead legacy code — those rows are today's activities and conditions, and both apps use them.

ID	WHERE	TASK
C2	Backend	A recording's activity or condition can be re-pointed at another company's. Attaching one is checked; changing it afterwards is not. Apply the same check on edit.
C3	Backend	The default for the whole API is still "anyone may call this", so any endpoint where a developer forgets to say otherwise is public. Make the default "must be signed in" and mark the genuinely public ones — sign-in, register, password reset.
C4	Backend	The backend answers to any address, and the full list of its endpoints is readable without signing in. Restrict both outside development.
C5	Backend	The statistics export answers differently for a recording that belongs to another company than for one that does not exist, which lets an outsider work out which recording numbers are real. Answer the same way in both cases.
C6	Backend	Sign-in treats any unexpected failure as "not signed in", so a database outage would quietly make every request anonymous instead of reporting a problem.

D · The phone and the backend disagree

Things the app sends that the backend does not accept. Each one fails quietly. These are the same items as in the mobile task list handed out with this document.

D1 The phone still uses the old address for labels and events HIGH MOBILE

The backend moved activities and labels to a new place and the web app followed. The phone still uses the old one — every label, category and event call still goes to `/categories/…` while the web goes to `/work-library/…`. It works today only because the old one has not been switched off; the day it is, every activity and label action on the phone stops.
Mobile is further along than it looks — statuses and the activity lists are already done, so this is a finishing job.

D2 Break markers never save HIGH MOBILE

Every break is recorded against label number 0, which does not exist on the backend, so the upload is refused every time. All of a session's events go up in one request, so a single break takes the rest of the session's events down with it.

ID	WHERE	TASK
D4	Mobile	A work-cycle step is allowed to have no duration — for instance one imported from a spreadsheet with the time column blank. The phone requires a number, so the whole step list fails to load and the operator cannot start recording at all.
D5	Mobile	Step timing is thrown away. The phone labels the recording type slightly differently from what the backend expects, so no step events are recognised and the per-step timing of the whole session is lost.
D7	Mobile	A category that sits inside another category crashes the labels screen. It only works today because none of the categories in use have a parent.
D8	Mobile	The phone can only ever create one kind of category — the type is hardcoded to WORK_TASK whatever the user picked — and the request succeeds, so the mistake is invisible.
D9	Mobile	The phone still reads a field the backend deleted, so anything that depended on it is quietly dead.
D10	Mobile	A stray character in the sensor details sent from lite mode ends up inside one of the stored values, so that sensor's details are wrong and its body part is left out of the session.

E · The system knows what went wrong and does not say

Every one of these is a support call. The information exists and is thrown away.

E1 Pages show a developer's note instead of the reason HIGH WEB

When a page fails to load, the customer is shown text a developer wrote for themselves — one screen literally reads "React query error thrown manually from ArchivedSessionsTable". The real reason from the server is captured and never shown. There are twenty-seven of these in the code today.

> **NOTE FOR YOU**
>
> Most sit on old pages that no longer render; roughly eight are still reachable. Depends on E2 — do them together, or the team builds against a shape that is about to change. This is the first of the web tasks handed out with this document.

E2 Anything a user gets wrong comes back as a server error HIGH BACKEND

Creating a recording, registering firmware and creating a company all report a duplicate value, a missing field and a genuine crash in exactly the same way, so no app can show anything useful. What the user can fix should say what to fix; only real failures should look like failures.

> **NOTE FOR YOU**
>
> The work-cycle import was the worst of these and is now fixed — it says which row and which column. Copy that shape to the rest.

E4 The error screen can lock up the browser tab HIGH WEB

When the app catches an unexpected error it calls its own reset while it is still drawing the error screen, which causes the error again, which resets again. It spins until the tab freezes. Both apps.

ID	WHERE	TASK
E3	Web	When the backend rejects a form and names the field, the client panel reads only the general message and shows a vague — sometimes empty — notification, with nothing marking the field. Put the message on the field it belongs to. The duplicate-email case now does this; nothing else does.
E6	Mobile / Web	Fields accept values the backend will refuse, so the user only finds out after submitting. On the phone: Worker ID, workplace, comment, age, weight, resting heart rate. On the web: company name, ticket subject, work-cycle step name.
E7	Web	Adding a factory with a name that already exists shows nothing on the field and the dialog just sits there. The team invite form also sends whatever is typed without checking it is a valid email address.
E8	Backend	Values that are too long reach the database and come back as an unreadable error.
E9	Backend	Updating several archive settings at once quietly skips any the user is not allowed to change and still reports success. Say what changed and what did not.
E10	Backend	Nothing stops two labels with the same name in one category, two operators with the same name on one workstation, or two report templates with the same name — so the user sees identical entries in a drop-down with no way to tell them apart.
E11	Web	"Try again" on the error screen only clears the error and draws the page again — it never asks the server for the data a second time, so the same error comes straight back. That screen is also hardcoded English in both apps: "Something went wrong" and "Try again" stay in English in every language.
E12	Mobile	The "no organisation" message is hardcoded English in two places — "Failed to create work session no Organization specified" — whatever the app language. Everything else is translated.
E13	Backend	Updating firmware that does not exist reports a validation problem instead of "not found".

F · Slow

The web items F1–F6, F8 and F9 are with the web team as *Web — speed cleanup* (17 August) and are not repeated here. What is left is the backend and the phone.

ID	WHERE	TASK
F7	Backend	Live-update features are wired to an in-memory channel layer, so they work in one server process only and fail silently the moment we run more than one.
F10	Mobile	The recordings list piles up background work as you scroll — each row starts several jobs and cleans up one, so a sync you started can keep running after you have left the screen.
F11	Mobile	The sensor clock is re-synced before every recording. There is a shortcut meant to skip it when it was done recently, and a units mistake means it never fires, so every recording starts slower than it needs to.
F12	Mobile	Switching between the two kinds of Bluetooth scan leaves the first one running. Starting the firmware scan cancels only the normal scan's listener and the reverse is also true, so the radio stays busy and stale devices show up in the new results.

G · Build, deploy and safety nets

G1 Broken code can be deployed DO FIRST WEB

The pipeline builds and ships the web apps from a step that never checks whether the type check, the linter, the build or the tests passed. Nothing stops broken code reaching customers. Make shipping depend on the checks passing.

> **NOTE FOR YOU**
>
> Top item on 10 July, top item on 27 July, top item on 12 August, still unchanged. Until this is done, anything fixed on the web side can come back the week after — and we have proof: two paid fixes were merged in May and silently undone by one merge in June, and nobody found out for eight weeks. The backend already has the safety net the web apps do not. This is the first item in the web task list handed out with this document.

ID	WHERE	TASK
G2	Web	High. The linter reports 430 problems, 52 of them errors, and the pipeline is told to carry on regardless so nobody sees them. Nearly half come from the admin app using a library it never declared — it works only because another app happens to install it. Clear them, declare the library, then let the linter stop a bad build.
G3	Web	High. The web apps still have exactly two test files and both assert that true is true. Start with the RULA and RAMP scoring — the number that tells a customer whether their worker is at risk. The backend went from nothing to a full suite in about a week this month; point the web devs at what they did.
G4	Web	High. Neither app is given the backend's address properly when it is built, so any image built the normal way has no valid address — and yet production works, which means something undocumented is supplying it and nobody knows what. Find out, wire it up, and fail the build loudly if it is missing. This is the difference between "we deploy" and "we can deploy again if the machine dies."
G5	Backend	Half fixed. The housekeeping job now runs every five hours instead of every minute. It is still started once per server process with nothing stopping duplicates, so with more than one worker it still races against itself.
G6	Web	A broken image build keeps the pipeline green — both builds are told to carry on after a failure, so a broken build and a good one look identical on the pull request.
G7	Web	The admin build ignores the record of which library versions we tested against, so two builds of the same code can install different software.

H · Cleanup and dead ends

Nothing here affects a customer today. Most are traps for the next developer.

ID	WHERE	TASK
H1	Web	A developer debugging panel is shipped to customers in both apps, and it is installed as a normal dependency rather than a development one, which is why nobody noticed.
H2	Web	The old v1 pages still exist behind switches that can no longer be turned on, and a few old pages are still reachable by typing the address. This is the single biggest source of confusion in the codebase — most of the wrong findings in earlier audits came from reading these dead screens. Remove them and retire those addresses, once we are sure v2 is not being rolled back.
H3	Web	Three places switch off type checking on data coming back from the backend, so a backend change breaks them for the user instead of failing the build. One of these caused the "Measured this month: 0" bug.
H4	Web	Dead admin interface: a bulk archive action left commented out, a removed notifications bell whose button would not have worked, and a "?" icon that looks pressable and only has a tooltip.
H5	Web	When a user exports a comparison they get a spreadsheet and a PDF that should share a name. The timestamp is worked out twice, so if the clock ticks between them the two files look unrelated — and around midnight they get different dates.
H6	Web	An unused file defines the "correct" names for cached assessment data while the whole app uses a misspelled version. Nothing is broken because everyone is wrong the same way — but the first developer who uses the correct file will silently break assessment refreshing, and it will look like an intermittent bug.
H8	Backend / Mobile	Worker records no longer have a first and last name, but both apps still carry them, the backend ignores them, and the phone splits the Worker ID into a pretend first and last name to display it. Nothing is lost — it is just confusing to read.
H9	Mobile	Switching company while offline reports success and never tells the server, so the user believes they are in the new company and their next recording is filed under the old one.
H10	Mobile	The first heart-rate reading after a pause measures its gap from before the pause, stretching it across the whole break and skewing the recovery figure.
H11	Mobile	A single malformed record crashes the whole work-cycle list instead of being skipped. The shared safety net only catches one of the two kinds of failure Dart has, and a bad reply from the server produces the other kind, so it goes straight past and leaves the screen stuck loading. Same for the company list and email sign-in.
H12	Web	Errors appear in the browser console on every page load. Harmless in themselves, but they bury real ones.
H13	Web	The prototype pages are open on the live site with no login. They touch no real data, but anyone with the address can look at them.
H14	Web	Plurals are hardcoded ("1 lines", "1 stations"), gender shows raw and inconsistently — "male" next to "MALE" in one column — and times under a second show as 00:00:00 next to a percentage that is not zero.
H15	Web	Comparison columns say "Session 1 / Session 2" instead of the measurement numbers, and body parts that were not recorded show as full tables of dashes instead of being hidden.
H16	Web	The measurement page shows one "View RAMP" link even when several assessments exist, so duplicates are invisible from there.
H17	Web	The Home to-do count and the notification bell count what has been loaded rather than the real total, so both stop being right past a few hundred items.
H18	Web / Backend	The Workers list still counts records that were created automatically for staff before that behaviour was removed. New staff no longer get one, but the old rows are indistinguishable from real workers, so this is a data clean-up rather than a filter.

Decisions for you

Not developer calls. Nothing here should become a ticket until it is answered. DEC-1 is closed — the code now says −5° for the head everywhere.

DEC-6 · ANSWER THIS FIRST

MEC question 11: what is it actually asking?

The screen calls it “Neck bending”. The code works it out as the head angle minus the trunk angle. The backend’s own internal name for it says it should be the head’s own angle. Three names, one calculation, and they do not agree.
The developer has asked whether to change the title instead of the calculation. That is your call. Decide what question 11 is measuring; the title and the code then both follow from it. B2 and B6 are both waiting on this.

DEC-7

What should happen when the AI section cannot be produced?

If the company has AI switched off, or the AI call fails, the report is generated without that section and nobody is told. The developer has asked how to handle it.
Two options: refuse the download and say why, or produce the report with a line where the section would have been saying it could not be generated. Pick one. B8 is waiting on this.

DEC-8

Which colour rule stays?

There are two separate pieces of code deciding the colour of a score and they disagree. A 1.5 comes out yellow on the compare-assessments table and green on the single results page — same assessment, same number, two screens, two colours. A score of 1 is green everywhere except question 1.8, where it is yellow, and that was written deliberately in July 2024.
Both rules have been found. Tell us which one is right and it gets applied everywhere.

DEC-2

Do gender and dominant arm affect any calculation?

The corruption is fixed — a recording no longer overwrites what an admin set, and the phone now sends the values in a form the backend accepts. What is still unanswered is whether those two fields feed any calculation. If they do, results from before August were affected and we need to know how far back.

DEC-3

How much is section C worth to us?

Nothing in section C affects a customer and there is no sign any of it has been used. Fixing it properly is real work. The counter-argument is that we sell to several companies and one reading another’s data ends a deal and starts a GDPR conversation. Fund it now, or schedule it.

DEC-4

Should two companies be allowed the same name?

Company names are unique across the whole system, so the second company called “Nordic Logistics” to sign up is refused. This may be deliberate. Answer it on purpose rather than by accident.

DEC-5

Self-signup: build it or delete the page?

There is a “Create an Account” page with a button that does nothing, and nothing in the product links to it. Given that email sign-in and referral registration both shipped recently, the real question is whether self-signup is on the roadmap at all.

Features, not bugs

FEATURE

An assessment you can save and come back to

A RULA, REBA or RAMP form is all-or-nothing today: fill in half of it, click away, and everything since the last Save is gone with no warning. There is no autosave and no draft.

The answer is a proper unfinished state the user controls — save what you have, see it marked as unfinished in the list, open it later and carry on. Design it together with A2, which is the same screen.

FEATURE

Choose how many rows you see

No list in the product lets you change the number of rows per page. Measurements is fixed at ten. If you want a control for it, it is new work — the old one is gone with the screens it lived on.

Parked on purpose

Real findings, deliberately not scheduled. Listed so nobody re-reports them.

- **Location stops being recorded after the first session** — it works for the first recording after the app opens and never again. You chose to hold this.
- **TNO analysis drops a whole middle range of angles** — for the trunk that is 0° to 20°, most normal working posture. Farhad: skip TNO for now.
- **Phone warnings versus report thresholds for the head** — Farhad wants a table comparing the two before deciding.
- **Four invitation-flow flaws** — the admin invite path emails a dead link, a failed email locks the invite for seven days with no resend, member limits are stored but never enforced, and deleting a pending invitee can delete the person entirely. You asked to hold these.
- **Five sensor-configuration problems** — the one that matters is a recording being processed at the wrong sample rate when the phone does not report which it used, producing wrong durations and angle statistics with nothing flagging it. You asked for no tickets.

Closed since 12 August

Re-read in the code today and confirmed done. Listed so they are not re-reported.

ID	WHERE	TASK
A3	Web	Deactivated team members now have a Reactivate action in the same menu Deactivate sits in.
A4	Backend / Web	Changing an email to one already in use now returns "This email is already in use" and the client panel shows it on the field.
A6	All three	Work-cycle steps have a saved order, the drag handle sets it, the backend sorts by it and the phone follows it.
A8	Mobile	The server's recording id is now written down before anything else can fail, so a sync no longer uploads a second copy.
A9	Backend	A work-cycle import is now all-or-nothing, and a bad row says which row and which column.
A10	Web	Work Library: the 5 MB limit is enforced, .csv/.xlsx/.xls are accepted properly, merging warns that it cannot be undone, and the retire warning shows how many tags go with it.
B1	Backend / Mobile	Head backward bending is −5° everywhere — risk verdict, Word report, RAMP and phone. Trunk stays −10°. This answers DEC-1.
B3	Backend / Web	A limb we did not measure no longer scores a green 0. A real zero shows as zero; a missing sensor shows as missing, with no score and no colour.
B5	Backend	Trunk and head forward/backward bending percentiles now print in the Word report. Side bending is left out, as decided.
B10	Web	RAMP item 1.5 asks its own question now, in all five languages.
B11	Web	RAMP sub-rows are named — forward, side and backward bending — instead of restarting at 1.1 inside every item.
B12	Backend / Web	Arm and forearm length are asked once per assessment and RAMP uses them, instead of assuming 32 cm and 27 cm for every worker.
D3	Mobile	Recorded body parts are sent in the form the backend stores, so they are no longer all reported as missing.
D6	Mobile	Choosing an operator now returns that operator's work cycles instead of the whole company's.
E5	Mobile	Opening the sensor scan with Bluetooth off now says so instead of spinning forever.

How this was checked

- All four repositories were pulled and read on `dev`: backend `00c6e35`, web apps `cb2d905d`, mobile `96aed6a6`, landing `bb99528` (unchanged since October 2024).
- Every commit made since the 12 August revision was read, and every item it could plausibly close was re-read in the code rather than taken from the commit message. Two claims made in the 18 August answers did not survive that check and are flagged at the top.
- Not re-read: A11 (admin forms), H3 to H6 and H12 to H18. These are small cleanup items that cannot be settled by reading code, and no commit has touched those files since 12 August, so nothing can have changed. Nothing on the list depends on them.
- The web speed items were removed from this list because they are already with the team as a separate document, not because they are done.